

Lecture 8: Climate Change II

ECON7720: Ecological and Environmental Economics

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Semester 2, 2026 | Required reading: Tietenberg & Lewis, Ch. 13

Recap of Lecture 7

- GHGs are **global, stock, uniformly mixed** pollutants with unique policy challenges.
- IPCC AR6: human-caused warming is **unequivocal**; tipping points and fat tails make risks asymmetric.
- Three broad strategies: sequestration, adaptation, **mitigation**.
- The **carbon budget** is finite: net zero by mid-century is required to limit warming to 1.5°C.
- The **social cost of carbon** depends critically on the discount rate: Stern (**low** r , act now) vs. Nordhaus (**higher** r , gradual action).

Lecture Outline

1. Carbon pricing: taxes vs cap-and-trade, hybrids, offsets, empirical evidence
2. Complementary policies: clean energy, leakage, revenue use, just transition
3. International environmental agreements: game theory, free-riding, climate clubs

Part 1: Carbon Pricing

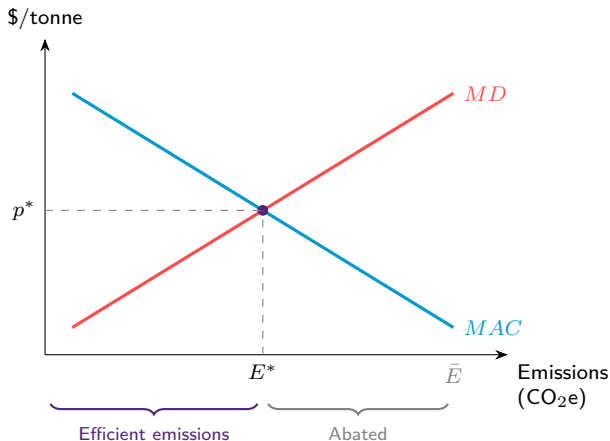
Why carbon pricing?

“A carbon price is the price for avoided or released carbon dioxide (CO₂) or CO₂-equivalent emissions. This may refer to the rate of a carbon tax, the price of emission permits, or the social cost of carbon.”

— IPCC AR6, Annex I: Glossary

- Carbon pricing is unique in its ability to control GHG emissions with a single instrument.
- Once a price is put on CO₂e, it promotes:
 1. Fuel switching to lower-carbon fuels.
 2. Energy efficiency by raising cost savings per unit of emissions reduced.
 3. Reductions in emissions from non-energy GHG sources.
 4. Technological innovation to develop new fuels and emission-reducing techniques.
- Cost-effective mitigation: *MAC* is equalised across all reduction opportunities when all sources face the same price per tonne CO₂e.

The efficient carbon price



- The **efficient carbon price** p^* equals both MD and MAC at the optimum E^* .
- Set $p^* = SCC \Rightarrow$ all emitters face the true social cost of their emissions.
- In practice, most carbon prices are **well below** $p^* \Rightarrow$ too much pollution.

Conditions for cost-effectiveness

- **Condition 1:** the carbon price should apply to **all** GHGs (via CO₂e) and **all** emission sources.
 - Otherwise, *MAC* is equalised only for the covered subset.
 - In practice: many programmes cover only some sources of some gases.
- **Condition 2:** all emitters should face the **same price** per tonne CO₂e.
 - Different prices $\Rightarrow$ different *MAC*'s $\Rightarrow$ violation of the equimarginal condition.
 - In practice: prices vary widely across programmes.

Bottom line

Carbon pricing *can* be fully cost-effective under the right conditions, but those conditions are **not yet commonly met** in practice.

Carbon taxes vs. cap-and-trade

Carbon Tax

Government sets the **price**.
Market determines the resulting
level of emissions.

Cap-and-Trade

Government sets the **quantity**
(cap on total emissions).
Market determines the *price*.

- For any allowed level of emissions, there is a price that achieves it, and vice versa.
- Historically: Scandinavian countries led with carbon taxes (1990s); US proposals leaned toward cap-and-trade. Today, **both** are widely used (e.g. the EU ETS is cap-and-trade).
- As of 2024 (World Bank): **75 carbon pricing instruments** across ~70 national and subnational jurisdictions, covering ~24% of global GHG emissions.
- ~74% of covered emissions under **ETS**; remainder under carbon taxes.

Carbon pricing: where do we stand globally?

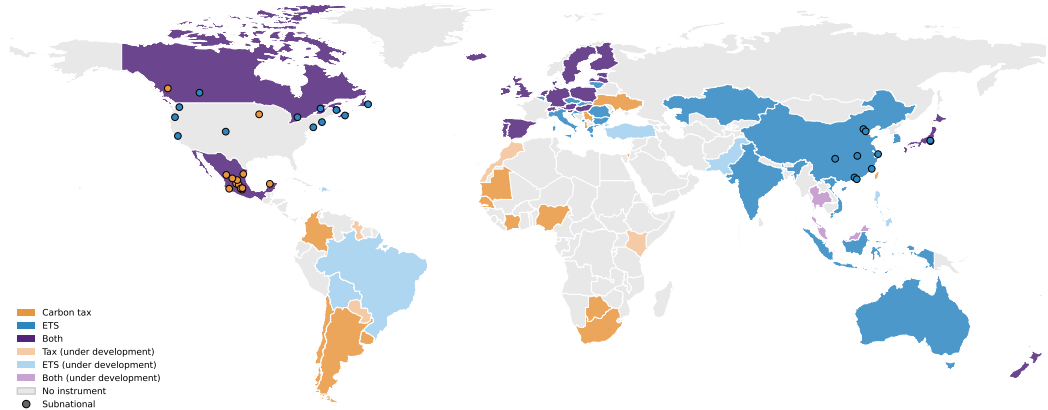
- Only ~24% of global GHG emissions are covered by a carbon price.
- Of those covered, fewer than 5% of emissions are priced at levels consistent with the Paris 2°C target (\$50–100/tonne by 2030, High-Level Commission on Carbon Prices, 2017).
- The **implementation gap** is enormous: even well-designed schemes cover only a fraction of the economy (e.g. EU ETS covers ~45% of EU emissions).

Recent momentum

- Number of carbon pricing instruments has tripled since 2010.
- New schemes: China ETS (2021), UK ETS (2021), India carbon market (under development).
- EU ETS prices reached record highs ($>€100$) in 2023.
- But: political setbacks too (Australia carbon tax repeal 2014, US lack of federal price, Canada moving to repeal federal carbon price 2025).

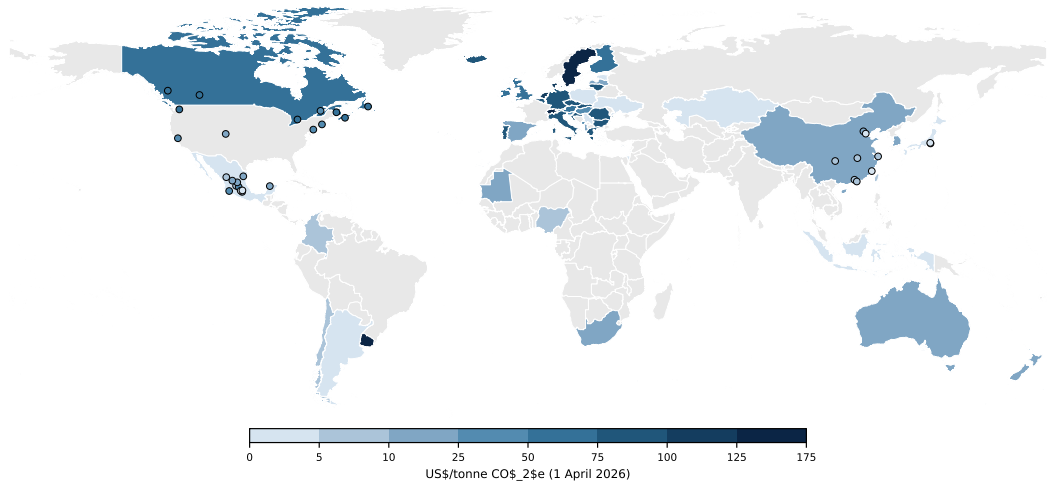
Note: Australia reintroduced carbon pricing via the reformed Safeguard Mechanism (2023): 220 facilities, baselines declining 4.9%/yr, Australian Carbon Credit Units (ACCUs) trading at A\$30–40/t; scheme under review 2026–27.

Carbon pricing around the world



75 carbon pricing instruments implemented (39 taxes, 36 ETSs) across 39 national and 33 sub-national jurisdictions, covering ~24% of global GHG emissions. Dots denote subnational instruments. Source: World Bank Carbon Pricing Dashboard (April 2026).

Carbon prices vary enormously



Prices range from \$0.7/t (Indonesia) to \$170/t (Norway). Only a handful of jurisdictions price carbon above the US EPA central SCC estimate of \$51/t. Most of the world's emissions remain un-priced or under-priced. For countries with multiple instruments, the highest national price is shown. Source: World Bank Carbon Pricing Dashboard, prices on 1 April 2026 (US\$/tCO₂e).

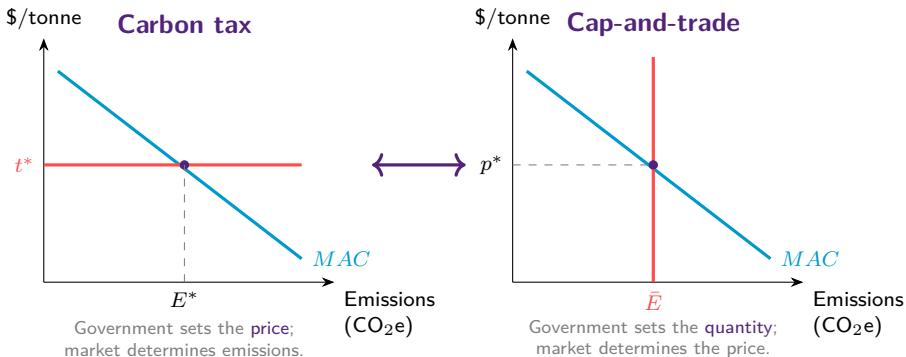
Comparing carbon taxes and cap-and-trade

Feature	Carbon Tax	Cap-and-Trade
Price certainty	Yes (set by govt)	No (market-determined)
Emissions certainty	No (market-determined)	Yes (set by cap)
Revenue generation	Direct & predictable	Depends on auction design
Admin. complexity	Lower	Higher (registry, monitoring)
Political framing	"Tax" (unpopular)	"Market" (more palatable?)
Price volatility	None (fixed rate)	Can be high
International linkage	Harder (tax harmonisation)	Easier (link ETS markets)

In practice, the distinction is blurring

- Cap-and-trade with price floors and ceilings approximates a carbon tax with a quantity backstop.
- Many jurisdictions use **hybrid** designs that combine elements of both.

Recall: price vs. quantity

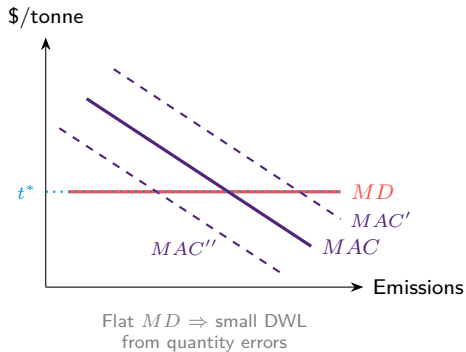


- With **perfect information**, a tax t^* and a cap $\bar{E}$ produce identical outcomes: same emissions, same price.
- The choice between them depends on **what we are uncertain about**, the shape of MAC or MD (Weitzman, 1974).

Price vs. quantity under uncertainty

Applying Weitzman (1974) to carbon pricing

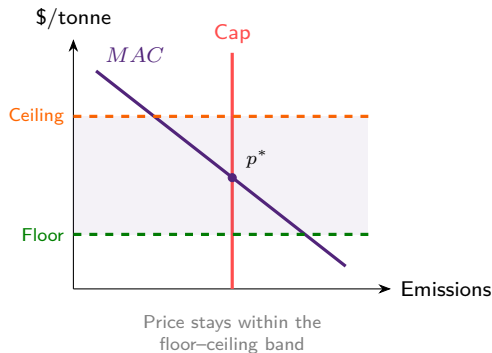
- GHGs are **stock pollutants** $\Rightarrow$ marginal damages are **flat** in the short run.
- Marginal abatement costs are **uncertain** (technology costs, economic conditions).
- Flat MD + uncertain MAC $\Rightarrow$ **price instruments** (carbon taxes) minimise welfare loss.



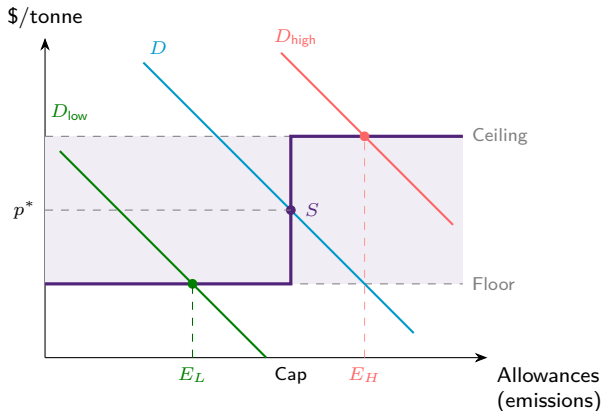
Hybrid designs: cap-and-trade with a price floor and ceiling

Why not just use a tax?

- Binding emissions targets (e.g. Paris Agreement) require **quantity certainty** that a pure tax cannot guarantee.
- Cap-and-trade provides **emissions certainty** but not cost certainty.
- **Hybrid designs**: cap-and-trade with a price floor and ceiling.
- Floor prevents price collapse; ceiling prevents spikes.
- Combines **environmental certainty** of a cap with **cost certainty** of a tax.



Hybrid designs: cap-and-trade with a price floor and ceiling



- **Normal demand** (D): price clears within the collar at the cap, standard cap-and-trade.
- **Low demand** (D_{low}): floor binds, allowances withdrawn (ECR), price cannot collapse. Emissions fall below cap ($E_L < \text{Cap}$).
- **High demand** (D_{high}): ceiling binds, extra allowances released (CCR). Emissions exceed cap ($E_H > \text{Cap}$).
- The collar creates a **hybrid**: quantity control within the band, price control at the boundaries.

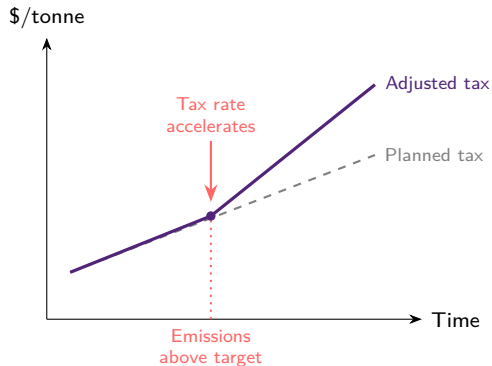
Hybrid designs: carbon tax adjustments

The problem

- A carbon tax fixes the **price** but not the **quantity**. Emissions may overshoot the target.

Automatic adjustment

- Track actual vs. target emissions (Metcalf, 2009, 2020).
- If emissions run **above** target $\Rightarrow$ tax rate escalates faster.
- If **below** $\Rightarrow$ tax rises more slowly.
- Acts like a thermostat: the tax **self-corrects**.



The symmetry of hybrid designs

- Price collar: makes cap-and-trade more like a tax (adding **price certainty**).
- Automatic adjustment: makes a tax more like a cap (adding **quantity feedback**).

Carbon offset markets (1)

What is a carbon offset?

A transferable instrument certifying an emission reduction of 1 tonne CO₂e from a project that reduces emissions below business-as-usual levels.

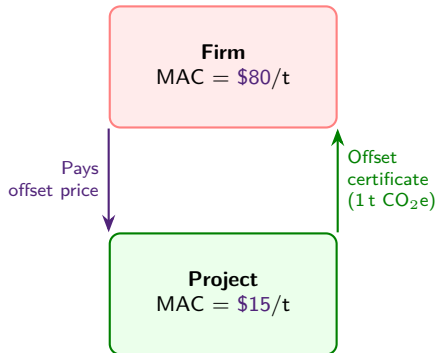
- Roles of offsets in carbon pricing:
 1. Lower compliance costs by expanding reduction projects.
 2. Increase the likelihood of enacting carbon pricing (easier compliance).
 3. Extend the reach to uncovered sources.
 4. Separate financing (buyer) from reduction (project investor).
- Markets can be voluntary or compliance-based.

Carbon offsets markets (2)

- Not every tonne of CO₂ costs the same to avoid: some reductions are **cheap** (e.g. methane capture), others are **expensive** (e.g. retrofitting a steel plant).
- Offsets let a firm with high abatement costs **pay someone else** to reduce more cheaply on its behalf.
- Same environmental outcome at **lower total cost** (cheapest abatement first).

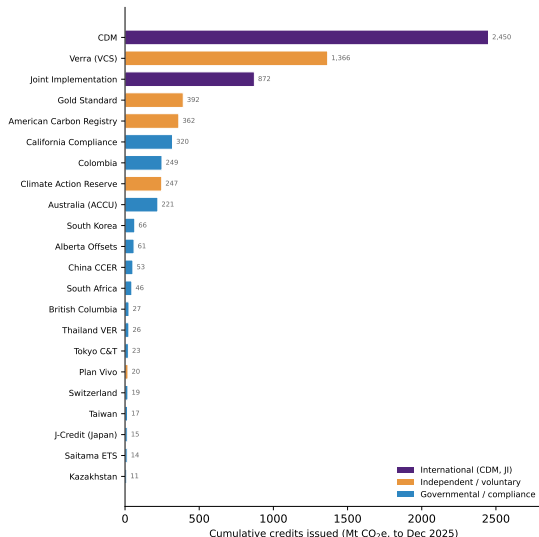
How do offsets differ from ETS?

- ETS: firms trade allowances **within** a capped sector; total emissions fixed by the cap.
- Offsets: credits from projects **outside** the capped system, expanding the pool of reduction opportunities.



Firm saves \$80 — offset price
Project earns offset price — \$15
Both are better off

Carbon offset markets (3)



Source: World Bank Carbon Pricing Dashboard (May 2026), cumulative credits issued to 31 December 2025.

Three types of crediting mechanism:

1. **International (purple)**: created under the Kyoto Protocol.

- CDM (Clean Development Mechanism): developed-country firms fund projects in developing countries.
- JI (Joint Implementation): projects between developed countries.

2. **Voluntary (orange)**: private standards, no regulatory mandate.

- Verra (VCS), Gold Standard, ACR: firms buy credits for corporate climate pledges.

3. **Compliance (blue)**: government-run, linked to domestic ETS or regulations.

- California, Australia (ACCU), South Korea, China (CCER).

Total: **6.5 Gt** CO₂e issued cumulatively. CDM dominates historically (2.5 Gt) but voluntary markets now lead annual issuance.

How has carbon pricing worked in practice?

1. Cost savings

- *Ex ante* analyses: shifting from command-and-control to market-based instruments achieves similar reductions at **much lower cost**.
- *Ex post* studies: cost savings are positive but **less than fully cost-effective**.
- Carbon pricing produces more emissions reduction per dollar than renewable subsidies or mandates.

2. Co-benefits

- Reducing CO₂ simultaneously reduces SO₂, NO_x, and particulates ⇒ significant **health benefits**.
- Stanford Energy Modeling Forum (EMF): expected economic costs are modest and offset by avoided climate damages and health benefits.
- Employment: job losses in carbon-intensive sectors largely **offset by gains** in other sectors.

Empirical evidence on carbon pricing

Andersson (2019, *AEJ: Economic Policy*)

Sweden's carbon tax (introduced 1991) reduced transport CO₂ emissions by $\approx 11\%$, using a synthetic-control design with other Nordic countries.

Metcalf & Stock (2023, *AEJ: Macroeconomics*)

Carbon taxes in 31 European countries reduced emissions by 4–6% on average within three years, with no significant negative effect on GDP or employment.

Känzig (forthcoming, *American Economic Review*)

EU ETS permit price shocks reduce emissions but have unequal consequences: low-income households bear a disproportionate cost burden, raising equity concerns for carbon pricing design.

Has carbon pricing met its goals?

World Bank, *State and Trends of Carbon Pricing* (2021)

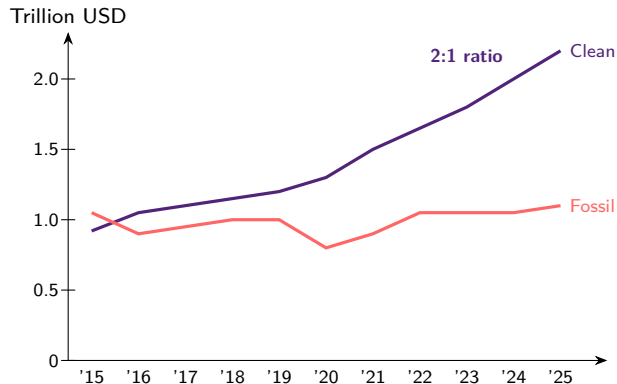
*“The potential of carbon pricing is still largely untapped, with most carbon prices **below the levels** needed to drive significant decarbonisation. [...] Global emissions have continued to rise and current climate policies [...] continue to fall far short of what is needed to reach the temperature goals of the Paris Agreement.”*

- Implemented prices are insufficiently aggressive in two senses:
 1. The **amount** of reduction is too small.
 2. Its **timing** is too slow.
- The political will has fallen short.
- **Complementary policies** are needed: renewable portfolio standards, investment tax credits, R&D support.

Complementary policies and R&D

- Carbon pricing alone is **not sufficient**, other market failures require other instruments.
- **R&D externality**: gains from innovation not fully captured by the innovator $\Rightarrow$ under-investment.
 - Public R&D funding has driven major innovations (internet, vaccines, solar panels).
 - But: risk of funding failures and crowding out private investment.
- **Network externalities**: e.g. electric vehicles need charging stations, but lack of EVs holds back station supply (and vice versa) (Stock, 2021).
 - Two stable equilibria: few EVs/few stations vs. many EVs/many stations.
 - Policy can help tip toward the “good” equilibrium.
- **Information failures**: consumers may not know the energy costs of appliances/buildings $\Rightarrow$ energy efficiency standards.

Clean energy investment has overtaken fossil fuels



Source: IEA World Energy Investment (2025).

- Clean energy investment reached **\$2.2 trillion** in 2025, double fossil fuel spending.
- Clean overtook fossil fuels in 2016, shortly after the Paris Agreement.
- Solar PV alone: >\$500 bn/yr (more than all other generation sources combined).
- China leads (~\$680 bn), followed by the EU (~\$370 bn) and US (~\$300 bn).

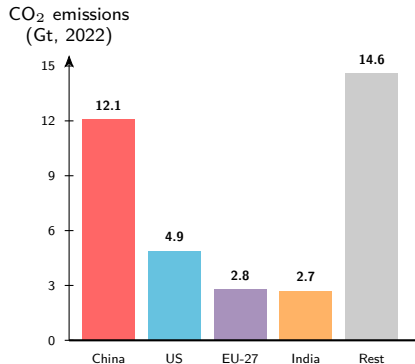
China's dual role

Largest emitter AND largest clean energy investor

- **Emitter:** ~31% of global CO₂ (2022); coal ~60% of electricity.
- **Clean energy leader:** largest producer of solar, wind, batteries, EVs (>50% of global EV sales).
- **NDC:** peak CO₂ before 2030; carbon neutrality by 2060.
- **ETS** launched 2021; planned expansion.

Why China matters

Without China, no climate agreement can succeed, but cooperation is complicated by geopolitics and development priorities.



Source: Global Carbon Project (2023).

The scale of clean energy investment

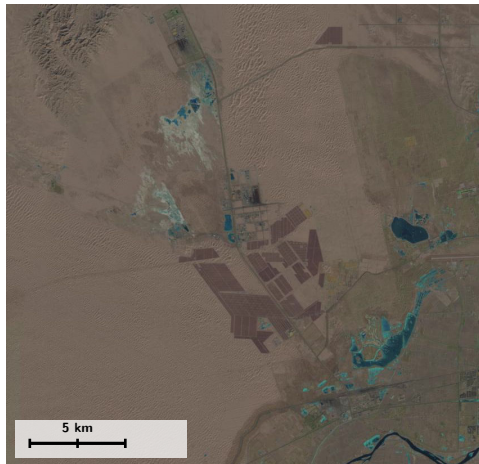
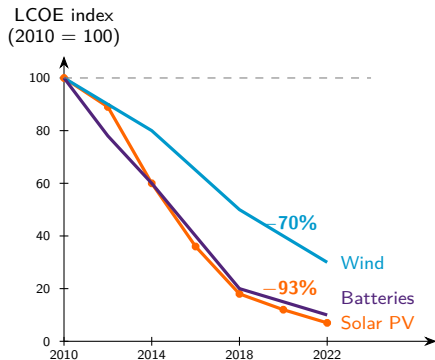


Image: Landsat 8 / USGS–NASA (public domain), via Wikimedia Commons.

- **Tengger Desert Solar Park**, Ningxia, China, one of the world's largest solar installations (1,547 MW, $\sim 43 \text{ km}^2$).
- China installed more solar capacity in 2023 alone ($\sim 217 \text{ GW}$) than the entire cumulative capacity of any other country.
- Global clean energy investment reached **\$1.8 trillion** in 2023, with China accounting for roughly one-third (IEA, 2024).

Renewable energy cost declines

- Solar PV costs fallen $>90\%$ since 2010; wind $\sim 70\%$; batteries $\sim 90\%$.
- New renewables now cheaper than new fossil fuels in most countries, even without subsidies.
- But: challenges remain in hard-to-abate sectors (heavy industry, aviation, shipping).



LCOE = levelised cost of energy. Indexed to 2010 = 100. Source: IRENA (2023), *Renewable Power Generation Costs*.

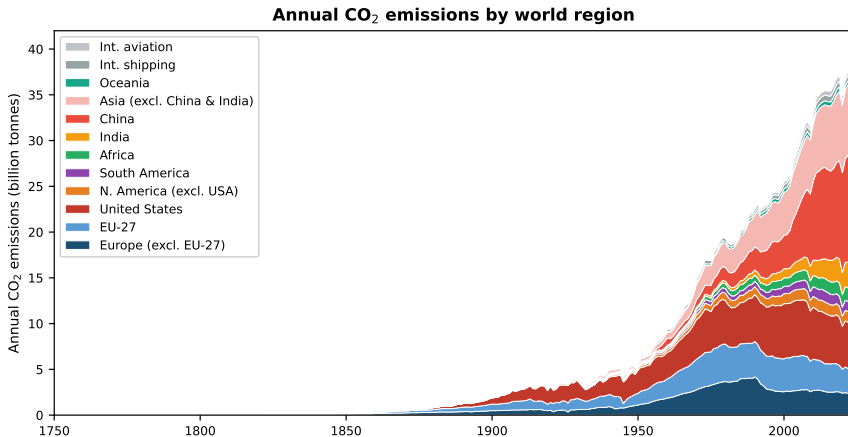
Carbon leakage

- Climate policies are not internationally uniform.
- Without a carbon price, foreign products are **cheaper** to produce $\Rightarrow$ more competitive.
- Firms may relocate carbon-intensive production to non-pricing jurisdictions $\Rightarrow$ emissions are **shifted, not reduced**.

The solution: border adjustment mechanisms

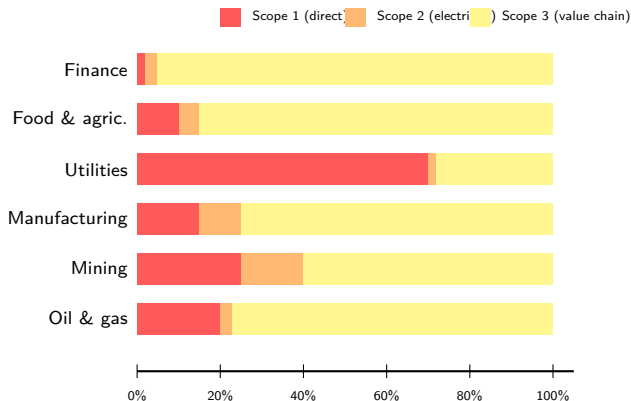
- Impose a **tariff** on carbon-intensive imports from jurisdictions with weaker or no carbon pricing.
- Tariff magnitude based on the **carbon intensity** of traded goods, seeking to equalise embedded carbon costs across countries.
- Eliminates the inefficient incentive for shifting emissions; tariff revenue retained by the higher-carbon-price country.
- Key example: the **EU CBAM**.

Carbon leakage: Global CO₂ emissions by region



Source: Global Carbon Budget (2025) via Our World in Data. CC BY. Emissions measured in billion tonnes of CO₂ per year from fossil fuels and industry; land-use change excluded. Global CO₂ emissions have risen from near zero in 1750 to over 37 billion tonnes per year. China and the United States are now the two largest emitters; Europe's share has declined since the 1970s. Without coordinated carbon pricing, emissions simply shift across borders, illustrating the carbon leakage problem.

Carbon accounting



- **Scope 1** — direct emissions from owned or controlled sources (combustion, vehicles).
- **Scope 2** — indirect emissions from purchased electricity, steam, heating.
- **Scope 3** — all other value-chain emissions (supply chain, product use, disposal).
- Scope 3 typically accounts for **>70%** of a company's total carbon footprint.

Scope definitions follow the GHG Protocol Corporate Standard (WRI & WBCSD, 2004; revised 2015). Sector averages are illustrative, based on CDP (2023) disclosure data.

Linking carbon markets internationally

Why link ETS markets?

- Linking two or more ETS creates a **larger market** $\Rightarrow$ more abatement opportunities $\Rightarrow$ lower costs (gains from trade).
- Prices converge across linked systems, promoting **cost-effectiveness** globally.

Challenges

- Different cap stringencies $\Rightarrow$ price arbitrage and potential **distributional** tensions.
- Linking requires harmonised **MRV** (monitoring, reporting, verification) standards.
- Political sovereignty concerns: linked partners lose some control over domestic carbon prices.

Article 6 of the Paris Agreement

Establishes rules for **international carbon market mechanisms**: cooperative approaches (Art. 6.2) and a centralised mechanism (Art. 6.4) to replace the Kyoto-era CDM, with safeguards against double counting.

Case study: Australia's carbon pricing journey

Year	Event
2011	Clean Energy Act: fixed carbon price ("carbon tax") of A\$23/tonne
2012–14	Price applied to ~500 largest emitters; emissions fell
2014	Repealed by Abbott government, only country to repeal a carbon price
2016	Safeguard Mechanism: baseline-and-credit for large emitters
2023	Reformed Safeguard Mechanism: declining baselines, tradeable credits

Lessons

- Carbon pricing can be **politically fragile**; design must anticipate distributional opposition.
- Revenue use (household compensation) was implemented but **political messaging** around "tax" was decisive.
- The reformed Safeguard Mechanism functions as a **baseline-and-credit ETS** for the ~215 largest facilities.

Revenue use and the double dividend

What to do with carbon pricing revenue?

1. Reduce the burden on low-income households and displaced workers.
2. Cut distortionary taxes (income, payroll) $\Rightarrow$ the “double dividend.”
3. Fund clean energy investment or public services.
4. Revenue neutrality: return all revenue via rebates (e.g. British Columbia, 2008).

The double dividend hypothesis

- First dividend: reduced environmental damages.
- Second dividend: using revenue to cut distortionary taxes improves economic efficiency.
- “Weak” form (revenue recycling beats lump-sum rebates): widely supported.
- “Strong” form (policy boosts GDP even ignoring environmental benefits): debated; depends on which taxes are replaced.
- British Columbia: evidence of a weak second dividend (GDP growth not harmed).

Goulder (1995, *J. Econ. Lit.*); Bovenberg & de Mooij (1994, *AER*); Metcalf (2019, *Brookings Pap. Econ. Act.*).

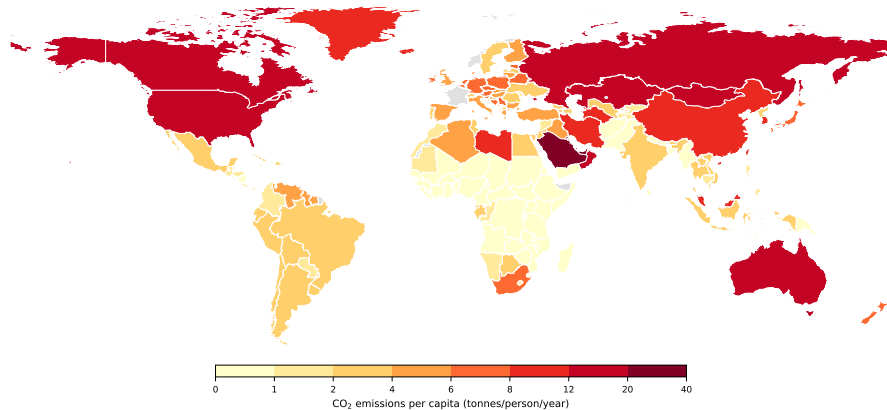
Distributional implications of carbon pricing (1)

- Climate **damages** fall disproportionately on lower-income households (both within and across countries).
- ⇒ Benefits of reducing damages are distributed **progressively** (disproportionately received by lower-income groups).
- But carbon pricing works by increasing **energy prices**:
 - Lower-income households spend a greater share of income on energy.
 - ⇒ Cost burden of carbon pricing is typically **regressive**.

The key to progressivity: revenue use

- Direct rebates to households (lump-sum dividends).
- Targeted transfers to vulnerable communities and workers.
- Investment in clean energy access in low-income areas.
- The design of revenue recycling determines whether the **overall** policy is progressive or regressive.

Distributional implications of carbon pricing (2)



Historical responsibility vs. **vulnerability**: high-income countries (dark red) emit far more per capita, yet climate costs fall disproportionately on the lowest emitters. The **Green Climate Fund** (2010) and **Loss and Damage Fund** (COP27, 2022) aim to bridge this gap. *Note*: CO₂ emissions per capita (tonnes/person/year, latest available). Source: Global Carbon Project via Our World in Data.

Carbon pricing and innovation

Does carbon pricing drive technological change?

- **Theory:** carbon pricing raises the return to emission-reducing innovation, encouraging R&D and adoption.
- **Evidence:** carbon pricing has spurred some innovation (e.g. EU ETS and low-carbon patents), but less than optimal.
- **Key barriers:**
 - **Knowledge spillovers:** innovators cannot capture all returns $\Rightarrow$ under-investment in R&D.
 - **Price uncertainty:** volatile carbon prices reduce incentives for long-term R&D commitments.
 - **Insufficient price levels:** current carbon prices are too low to justify costly breakthrough technologies.

Policy implication

Carbon pricing provides a necessary **demand pull** for clean technology, but **technology push** policies (public R&D, subsidies for demonstration projects) are also needed. Neither alone is sufficient.

Empirical evidence on carbon pricing and innovation

Popp (2002, *American Economic Review*)

Higher energy prices induce energy-saving patents. Using US patent data, a doubling of energy prices increases patenting by 6%.

Acemoglu, Aghion, Bursztyn & Hemous (2012, *AER*)

Directed technical change model: carbon taxes **redirect** innovation toward clean technologies. Optimal policy combines carbon pricing *with* R&D subsidies.

Calel & Dechezleprêtre (2016, *Review of Economics and Statistics*)

EU ETS increased low-carbon patenting by regulated firms by $\sim 10\%$, but the effect was modest relative to what is needed.

Aghion, Dechezleprêtre, Hemous, Martin & Van Reenen (2016, *JPE*)

Auto industry: firms facing higher fuel prices patent more in **clean** technologies. Strong **path dependence**, early direction of innovation persists.

Carbon pricing: political economy challenges

Why is it so hard to implement?

- **Concentrated losses, diffuse benefits:** fossil fuel industries and energy-intensive firms face immediate costs; benefits are global and long-term.
- **Salience:** carbon prices make energy cost increases *visible*, even if overall welfare improves.
- **Distributional concerns:** regressive impact on lower-income households without careful revenue recycling.
- **Competitiveness fears:** industries worry about disadvantage vs. unpriced competitors (carbon leakage).
- **Ideological opposition:** resistance to “taxes” or “government intervention” in markets.

Overcoming resistance

Successful programmes (BC, RGGI, EU ETS Phase 4) combine **transparent revenue use**, **gradual price escalation**, **border adjustments**, and **stakeholder engagement**.

Key design principles for carbon pricing

1. **Broad coverage:** include as many GHGs and sources as feasible to equalise MAC across the economy.
2. **Uniform price:** all emitters face the same \$/tonne CO₂e; avoid exemptions that erode cost-effectiveness.
3. **Predictable price trajectory:** rising over time, signalled well in advance, to guide long-term investment decisions.
4. **Hybrid designs:** use price floors and ceilings to manage political and economic uncertainty.
5. **Border adjustments:** prevent carbon leakage to non-pricing jurisdictions.
6. **Progressive revenue recycling:** protect low-income households; invest in clean energy and R&D.
7. **Complementary policies:** address market failures not solved by carbon pricing alone (R&D, networks, information).
8. **Adaptive governance:** build in review and adjustment mechanisms (act–learn–act).

Part 2: International Environmental Agreements

The IPCC and Sixth Assessment Report (AR6)

- Intergovernmental Panel on Climate Change (IPCC, est. 1988): 195 member countries; the authoritative scientific body on climate change.
- AR6 (2021–23): hundreds of scientists, three working groups.

Projected impacts

- More intense **heatwaves**, drought, and coastal flooding.
- Ocean acidification, permafrost thaw, glacier loss.
- Health risks: heat stress, disease spread, **food insecurity**.
- Disproportionate impacts on **developing countries** and vulnerable populations.

Emissions pathways

- Limiting warming to **1.5°C** requires global CO₂ emissions to reach **net zero** by ~2050.
- Limiting to **2°C** requires net zero by ~2070.
- Current policies put us on track for **2.5–2.9°C** by 2100.

Recommended: IPCC AR6 Synthesis Report (2023), Summary for Policymakers.

Timeline of international climate agreements

Year	Agreement / milestone
1988	IPCC established
1992	UNFCCC (UN Framework Convention on Climate Change) signed at Rio
1997	Kyoto Protocol : binding targets for developed countries
2005	Kyoto enters into force; EU ETS launched
2009	Copenhagen Accord (COP15): no binding post-Kyoto agreement reached
2015	Paris Agreement : below 2°C / pursue 1.5°C; NDCs
2016	Kigali Amendment: HFC phase-down
2021	Glasgow (COP26): phase down coal, methane pledge
2022	Sharm el-Sheikh (COP27): Loss & Damage fund agreed
2023	Dubai (COP28): "transition away from fossil fuels"

Climate as a global public good

Applying the public goods framework from Lecture 2

- Climate stability is **non-rival**: one country's enjoyment of a stable climate does not reduce another's.
- Climate stability is **non-excludable**: no country can be excluded from the benefits of GHG reductions, regardless of whether it contributed.
- $\Rightarrow$ Climate mitigation is a **pure public good**.
- Classic problem: **under-provision** because each agent has an incentive to free-ride.
- Unlike local public goods, climate requires **global** cooperation, there is no government to provide the optimal level.

	Excludable	Non-excludable
Rival	Private goods	Common pool
Non-rival	Club goods	Public goods

$\uparrow$
Climate stability

The challenge of international cooperation

- Climate change mitigation is a **global public good**.
- Each country bears *all* costs of its reductions but gains only a *portion* of the benefits.
- Individual nations therefore have an incentive to **free-ride** on others' abatement efforts.
- Private marginal benefit < social marginal benefit $\Rightarrow$ too little collective reduction.

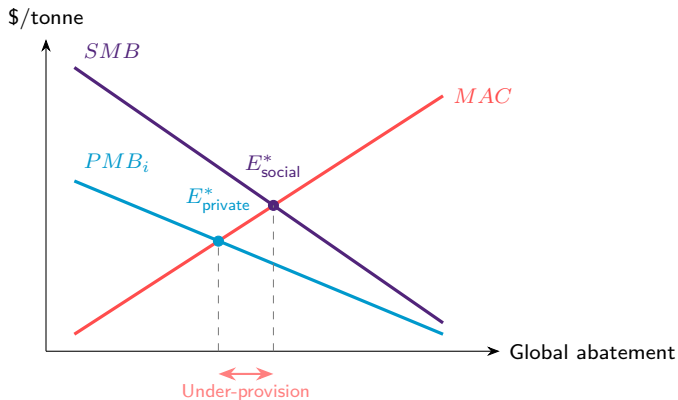
Requirements for success

- Agreements must be **self-enforcing**: no supranational authority can force compliance.
- Signatories can always **withdraw** (cf. US withdrawal from Paris, 2017; re-joined 2021; withdrew again 2025).



Delegates at the COP21 opening plenary, Paris, November 2015. With ~ 195 parties, each country captures only $\sim 0.5\%$ of the global benefit of its own abatement. Source: Presidencia de la República Mexicana via Wikimedia Commons (CC BY 2.0).

The free-rider problem: a graphical view



- Each nation's **private marginal benefit** (PMB_i) from abatement is less than the **social marginal benefit** (SMB), which is the sum across all nations.
- Nations abate where $PMB_i = MAC \Rightarrow$ too little global abatement.
- The gap between social and private optima is the **free-rider shortfall**.

Game theory and climate negotiations

The Barrett model of self-enforcing agreements

- When a country **joins**: other signatories respond by increasing their abatement (reward).
- When a country **withdraws**: remaining signatories reduce abatement (punishment).
- Key result: a stable, successful agreement can only be achieved when the difference between cooperative and non-cooperative outcomes is **small**.

The paradox

- In precisely those cases where **gains from cooperation are greatest**, the incentive to free-ride is also greatest $\Rightarrow$ inhibits stable agreements.
- Catastrophe thresholds (tipping points) can help *if* the specifics are known with certainty and benefits of avoidance are high relative to costs (Barrett, 2013).
- Uncertainty about tipping point temperatures normally causes coordination to **collapse**.

The N-country free-rider problem

- Suppose N identical countries each choose an abatement level.
- Benefits of abatement are shared **equally** by all N countries (global public good).
- Costs are borne **individually** by the abating country.
- Each country's **private** marginal benefit from its own abatement = $\frac{1}{N}$ of the social marginal benefit.
- As N increases, the private incentive to abate **decreases** $\Rightarrow$ more severe under-provision.
- With 195 countries: each bears 100% of its costs but captures only $\sim 0.5\%$ of the benefits.

Barrett's key result

The maximum size of a **stable** coalition is typically **small** when gains from cooperation are large. Only when gains are small does a large coalition become stable, but then there is little to gain from the agreement.

Nash equilibrium in the climate game

Why individual rationality leads to collective failure

- In Nash equilibrium, each country abates where its **private** marginal benefit equals its marginal cost:

$$C'(q_i^{NE}) = b$$

- But the **social** marginal benefit is $N \cdot b$, so the efficient condition is:

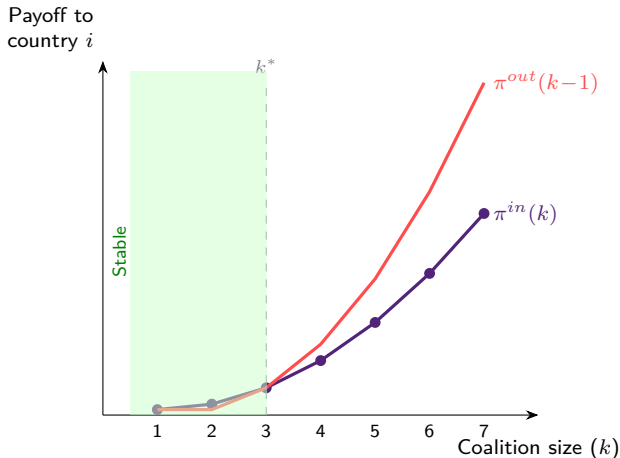
$$C'(q_i^*) = N \cdot b$$

- Result: $q_i^{NE} < q_i^*$ —each country under-abates by a factor related to N .
- Total Nash abatement $\approx \frac{1}{N}$ of the social optimum (with quadratic costs and linear benefits).

The Prisoner's Dilemma

With ~ 195 countries in climate negotiations, the free-rider incentive is **enormous**. Each country captures only $\sim 1/195$ th of the global benefit of its own abatement. This is the fundamental obstacle to voluntary international cooperation.

Barrett's stable coalition size



- $\pi^{in}(k)$: payoff from **joining** a coalition of k .
- $\pi^{out}(k-1)$: payoff from **staying out** when $k-1$ others cooperate.
- **Internal stability**: no member wants to leave ($\pi^{in} \geq \pi^{out}$).
- The stable coalition k^* is **small**.
- Beyond k^* , the temptation to free-ride exceeds the benefit of membership.
- This is Barrett's paradox: cooperation is most needed when it is hardest to sustain.

Game theory: a numerical example (1)

Setup

Two countries; each can choose Low or High emissions.

Gross benefits depend on both countries' choices.

Control costs: Country A = \$3/yr, Country B = \$6/yr.

Gross Benefits		
	B: Low emissions	B: High emissions
A: Low emissions	(4, 8)	(1, 4)
A: High emissions	(2, 1)	(-1, -1)

Net benefits (gross benefits – control costs if Low)

	B: Low	B: High
A: Low	(1, 2)	(-2, 4)
A: High	(2, -5)	(-1, -1)

A pays \$3 when Low; B pays \$6 when Low. No cost if High.

Game theory: a numerical example (2)

Finding the Nash equilibrium

	B: Low	B: High
A: Low	(1, 2)	(-2, 4)
A: High	(2, -5)	(-1, -1)

- **B's perspective:** if A plays Low, B gets 2 (Low) vs. 4 (High) $\Rightarrow$ High. If A plays High, B gets -5 (Low) vs. -1 (High) $\Rightarrow$ High. **B's dominant strategy: High.**
- **A's perspective:** given B plays High, A gets -2 (Low) vs. -1 (High) $\Rightarrow$ High.
- **Nash equilibrium:** (High, High) = (-1, -1).

The tragedy

The cooperative outcome (Low, Low) = (1, 2) is **Pareto superior** to Nash, but neither country has an individual incentive to cooperate. This is a **Prisoner's Dilemma** applied to climate.

Game theory: the role of technology and side payments

Technological change

Suppose innovation reduces Country A's control cost from \$3 to \$1/yr.

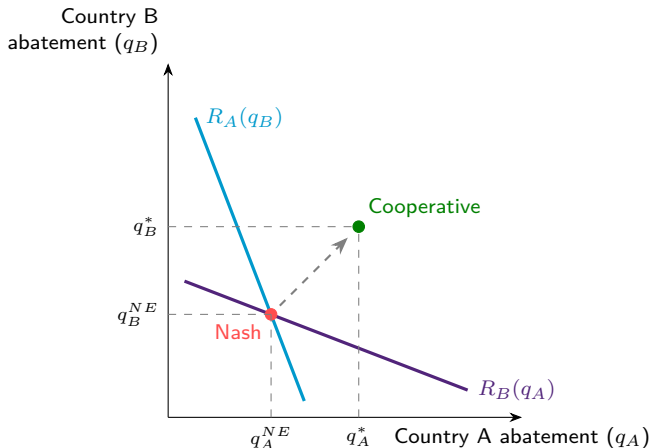
	B: Low	B: High
A: Low	(3, 2)	(0, 4)
A: High	(2, -5)	(-1, -1)

- Now A prefers Low emissions regardless of B's choice (dominant strategy).
- But B still prefers High (free-riding on A).
- New Nash equilibrium: (Low, High) = (0, 4).
- **Side payment:** if A pays B between \$3 and \$4, both choose Low emissions and both are better off.

Insight

Technology (lowering control costs) and **transfers** (side payments) can transform the game and make cooperation self-enforcing.

Best response functions



- Each country's **best response** to the other's abatement.
- Abatement is a **strategic substitute**: if A abates more, B free-rides and abates less.
- Nash equilibrium: low abatement by both.
- Cooperative optimum: both countries abate more than their best response, but each has an incentive to **deviate** back, so cooperation requires enforcement or repeated interaction.

Additional barriers to cooperation

Jacquet et al. (2013): timing of payoffs

- Lab experiments reveal that when payoffs from cooperation occur **with a lag**, it is more difficult to reach successful agreements.
- Climate payoffs are inherently delayed (decades to centuries) $\Rightarrow$ cooperation is harder.

Other challenges

- **Heterogeneity**: countries differ vastly in income, vulnerability, and emissions, makes agreement on burden-sharing very difficult.
- **Sovereignty**: countries resist binding commitments that constrain domestic policy choices.
- **Short electoral cycles**: politicians discount the future more heavily than social optimality requires.
- **Fossil fuel lobbying**: concentrated interests resist diffuse benefits.

Montreal vs. Paris: why climate is harder

Dimension	Montreal Protocol	Paris Agreement
Pollutant scope	Few chemicals (CFCs)	All GHGs, all sectors
Number of sources	Small (# of firms)	Billions (every person)
Substitutes	Available, profitable	Costly energy transition
Cost-benefit ratio	Clearly positive	Uncertain, long lag
Free-rider incentive	Low	High
Time horizon	Decades	Centuries
Enforcement	Trade penalties	Voluntary NDCs (Nationally Determined Contributions)
Distributional conflict	Manageable	Acute (rich vs. poor)

Key lesson

The conditions that made the Montreal Protocol successful (clear net benefits, few producers, available substitutes, strong enforcement) are **largely absent** for climate change, making cooperation far more difficult. But the Protocol shows international cooperation *is* possible when incentives are aligned.

Pathways to cooperation

Strategies that improve the odds of agreement

- **Co-benefits:** reducing GHGs also reduces local air pollution (e.g. particulates from coal). Health benefits are received by the mitigating nation $\Rightarrow$ no free-rider effect on that portion.
- **Issue linkage:** simultaneously negotiating trade liberalisation, R&D cooperation, or debt reduction alongside climate agreements.
- **Financial transfers:** high-income countries have a stake in helping low-income countries reduce emissions (all nations benefit from mitigation anywhere).
- **Cost-effective policies:** lower costs $\Rightarrow$ higher net benefits $\Rightarrow$ more likely participation.
- **Repeated interaction:** climate negotiations are ongoing (annual COPs) $\Rightarrow$ reputation effects and reciprocity can support cooperation.

COP milestones

COP	Year	Key outcome
COP1	1995	Berlin Mandate: launched Kyoto negotiations
COP3	1997	Kyoto Protocol adopted
COP15	2009	Copenhagen Accord: failed binding treaty
COP21	2015	Paris Agreement adopted
COP24	2018	Katowice: Paris rulebook agreed
COP26	2021	Glasgow: phase-down coal; Global Methane Pledge
COP27	2022	Sharm el-Sheikh: Loss & Damage fund
COP28	2023	Dubai: “transition away from fossil fuels”

- Annual COPs provide a repeated-game structure that can support reputation effects and gradual trust-building.
- But: the consensus rule means any single country can block strong language (e.g. Saudi Arabia, Russia weakening fossil fuel phase-out commitments).
- “Transition away from fossil fuels” at COP28 was the first time fossil fuels were mentioned in a COP decision text.

The Green Climate Fund

Financing climate action in developing countries

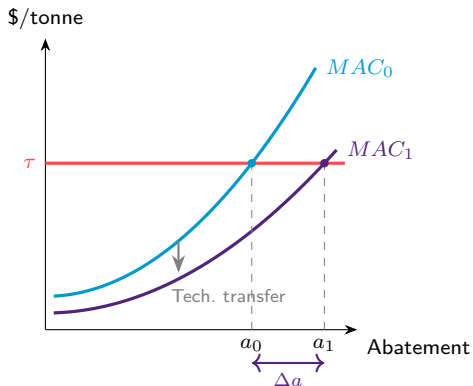
- Established in 2010 under the UNFCCC; headquartered in Songdo, South Korea.
- Purpose: support developing countries in **mitigation** and **adaptation** efforts, focusing on the most vulnerable nations.
- Developed countries pledged **US\$100 billion/year** by 2020, a target not met until 2022 (and disputed by developing nations).
- GCF has approved >US\$12 billion in projects since 2015 (renewable energy, resilience, forest protection).

Loss and Damage Fund (COP27, 2022)

- Addresses damages that go **beyond adaptation** (e.g. permanent land loss from sea level rise, destroyed infrastructure from extreme events).
- Initial pledges: ~US\$700 million (far below estimated needs of \$400+ billion/year).
- A symbolic breakthrough: developed countries acknowledged **responsibility** for climate impacts for the first time.

Technology transfer and capacity building

- Developing countries often lack **technology**, **finance**, and **institutional capacity** to reduce emissions.
- Key mechanisms: **Green Climate Fund**, UNFCCC Technology Mechanism, bilateral aid (Paris Agreement, Art. 10).
- Technology transfer shifts the MAC curve down $\Rightarrow$ more abatement becomes affordable at any carbon price τ .
- Renewable costs have fallen $\sim 90\%$ since 2010, making clean energy the **cheapest** option in most countries.
- Key debate: **intellectual property** rights vs. free technology transfer.



Falling costs shift MAC down $\Rightarrow$ more abatement at any carbon price τ .

Climate clubs and border adjustments

The “climate club” idea (Nordhaus, 2015, 2021)

- Two policy parameters:
 1. A stipulated **international carbon price** that all club members enact.
 2. A uniform **tariff** on all goods exported from non-members to club members.
- Nordhaus (2021) finds this might yield a stable cooperative solution, especially with high rates of technical progress.
- The tariff creates a **cost of non-membership**, reducing the incentive to free-ride.

EU Carbon Border Adjustment Mechanism (CBAM)

- Announced 2021; transitional phase from 2023; fully operational from **2026**.
- Covers imports of steel, aluminium, cement, fertilisers, electricity, hydrogen.
- Importers must purchase **CBAM certificates** based on embedded carbon, priced at EU ETS levels.
- If a carbon price was already paid in the exporting country, the amount is deducted.

Trade and climate: WTO compatibility

- **Carbon leakage:** production relocates to countries without carbon prices.
- **CBAMs** charge importers for embedded carbon, levelling the playing field.
- WTO compatible *if*: (i) mirrors domestic carbon costs, (ii) credits foreign carbon prices, (iii) non-discriminatory (GATT Art. XX exceptions).
- Controversy: the UNFCCC principle of “common but **differentiated responsibilities**” holds that rich countries (historically the largest emitters) should bear greater costs. Developing countries argue CBAMs reverse this by taxing *their* exports for a problem they did not create.

EU CBAM covered sectors

Aluminium	Steel
Cement	Fertilizers
Electricity	Hydrogen

Most exposed:
small economies with concentrated
exports to EU (e.g. Mozambique,
Ukraine, Egypt)

Source: World Bank CBAM Exposure Index (2023).

Appendix

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